U.S. Health Care Cost & Utilization: A SQL Portfolio Project

Who spends and who pays for care, and how much hospital prices vary for the same diagnosis, using MEPS 2024 and CMS Medicare Inpatient 2024.

MEPS HC-256 (2024) | CMS Medicare Inpatient (2024) | Compiled 25 Oct 2026

Summary

Ten percent of Americans account for 65.6% of health spending, and hospital list prices are 5.0 times what Medicare-covered stays actually pay.

The project asks who uses and pays for U.S. health care and how prices vary, using two public data sets for 2024: the AHRQ Medical Expenditure Panel Survey (MEPS) and the CMS Medicare Inpatient Hospitals file. The analysis is 13 SQL scripts in SQLite plus a Python script for survey standard errors.

65.6%
of spending is in the top 10% of people (2021 benchmark: 66.9%). The bottom half holds 2.8%.

$28,077
mean spending per adult with 5+ chronic conditions, against $4,046 with none.

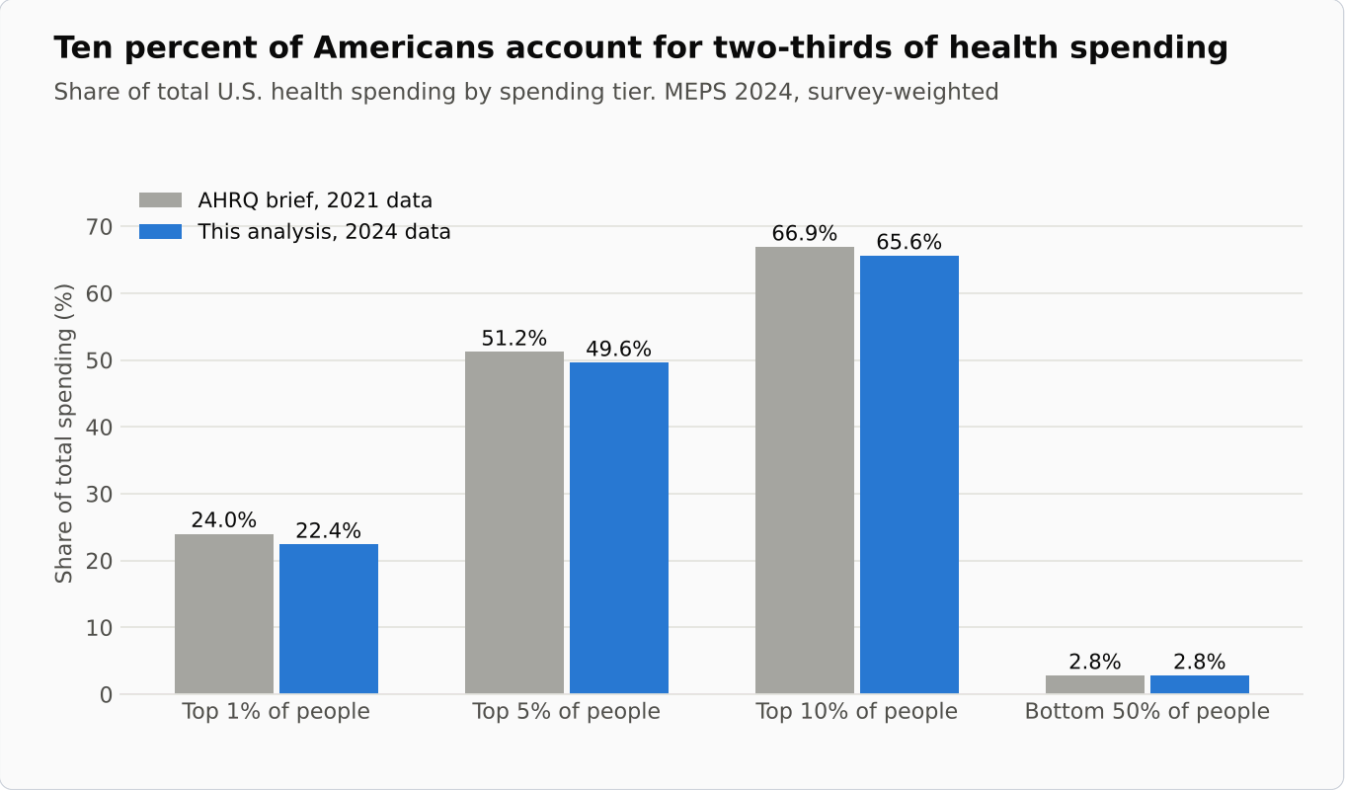
8.8
ER visits per 100 uninsured people, against 15.9 private and 31.9 public-only. The uninsured use the ER least.

5.0×
charges over payment: $92,408 billed against $18,360 paid per Medicare stay.

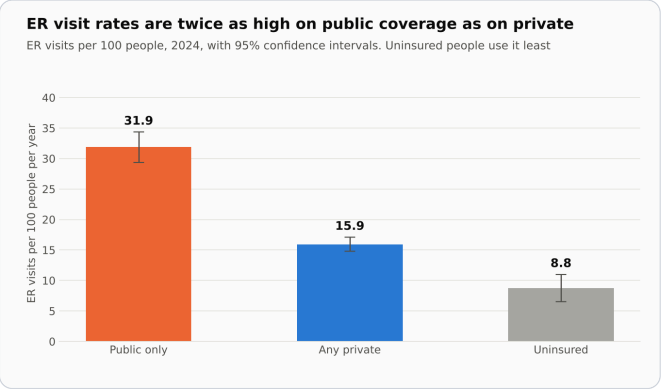
3.5× vs 15%
The highest-charging quarter of hospitals charges 3.5× the lowest quarter but is paid only 15% more.

4.4× vs 1.8×
Sepsis (DRG 871): the 90th-percentile hospital charges 4.4× the 10th percentile but is paid 1.8×.

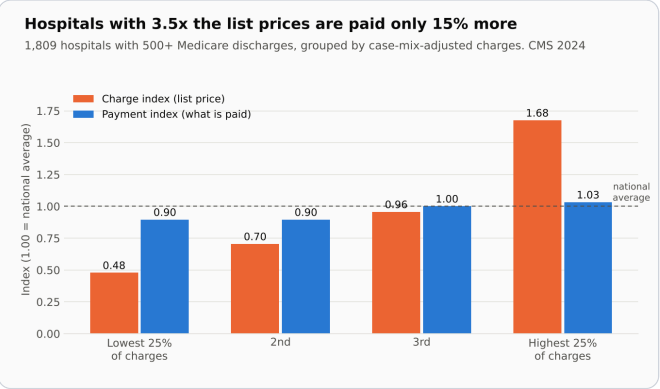
Charts



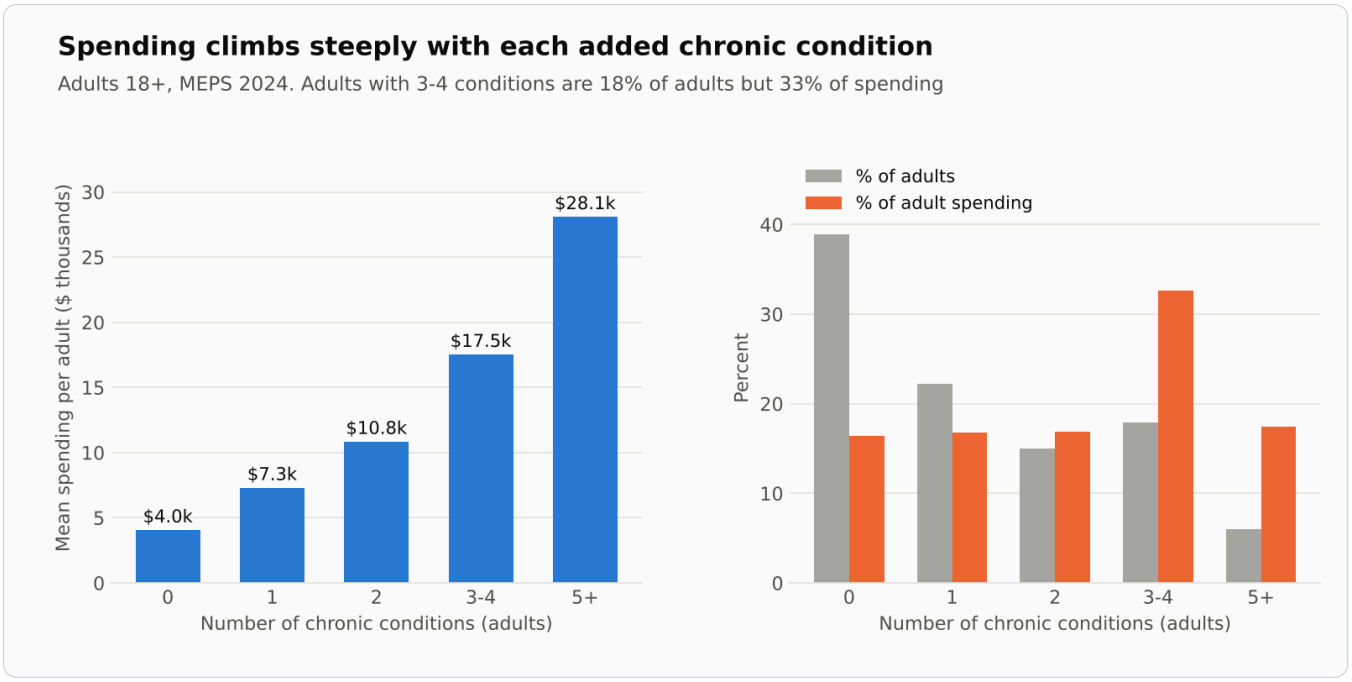
Spending concentration, MEPS 2024 against the published 2021 AHRQ figures.



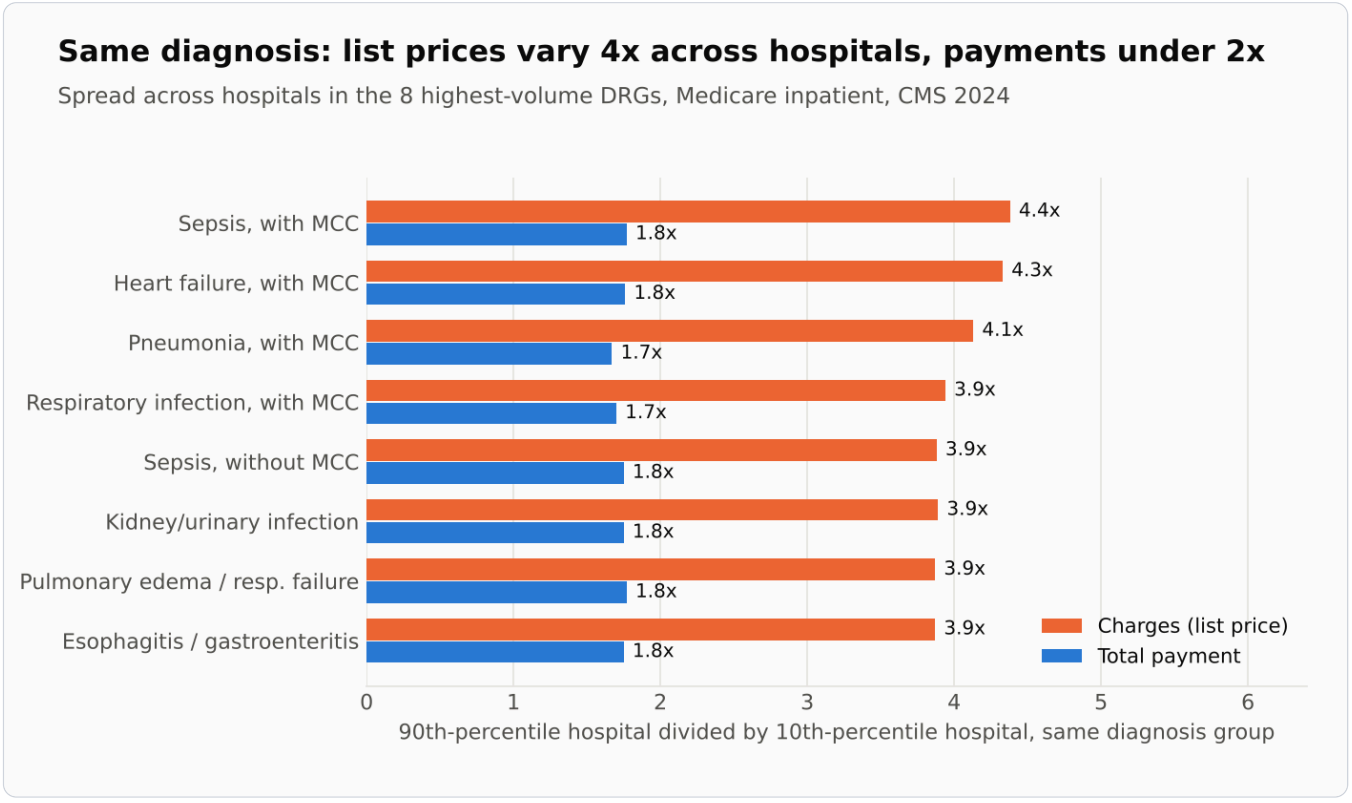
ER visits per 100 people by insurance, with 95% confidence intervals.



Charges climb 3.5-fold across hospital quartiles; payment rises 15%.



Spending by number of chronic conditions, adults 18+, MEPS 2024.



Spread across hospitals within the same diagnosis group, Medicare inpatient, CMS 2024.

Data and method

The two sources answer different halves of the question and cannot be joined row by row, so they sit side by side in one database.

SOURCE	UNIT	SIZE	COVERS
MEPS HC-256, 2024	one person	18,683 people with a survey weight	Civilian noninstitutionalized U.S. population, all ages and insurance types
CMS Medicare Inpatient by Provider and Service, 2024	one hospital and diagnosis group (DRG)	145,879 rows, 2,906 hospitals, 540 DRGs, 4.95M discharges	Medicare fee-for-service hospital stays only

- **Survey weights.** Every MEPS estimate uses the person weight, and spending tiers are cut on cumulative weight rather than row counts.
- **Design-based uncertainty.** SQLite cannot compute survey variance, so a Python script uses Taylor-series linearization with the survey's strata and sampling units. Its point estimates match the SQL.
- **Discharge-weighted CMS averages.** Each CMS row is a hospital-DRG average, so every rollup weights by discharges.
- **Case-mix adjustment.** Hospital and state indexes divide actual payment (or charges) by what the same cases would receive at the national average for each DRG. Only hospitals with 500 or more discharges are ranked.

Benchmark check: weighted 2024 concentration (top 1% 22.4%, top 5% 49.6%, top 10% 65.6%) lands within 1.6 points of the published 2021 figures at each tier.

MEPS findings: who spends and who pays

The costliest 5% of people are older (mean age 57, 41% aged 65+), almost all insured (0.3% uninsured), and 73.5% of the adults among them have two or more chronic conditions. Their spending is 28% inpatient, 29% prescriptions and 19% office visits. People 65+ spend $15,738 a year, 1.9 times the average, and account for 35% of all spending. Across all ages, private insurance pays 43% of spending, Medicare 26%, Medicaid 13% and households 13% out of pocket.

MEASURE	PUBLIC ONLY	ANY PRIVATE	UNINSURED
Share of people	28.9%	64.7%	6.5%
Mean spending per person	$9,908	$8,375	$1,099
ER visits per 100 people	31.9	15.9	8.8
Share with 1+ ER visit	20.6%	11.7%	7.7%

The public-only versus private spending gap ($1,533, standard error about $576) is larger than sampling error, but it is unadjusted: public-only enrollees are older and more often disabled, so it says who is on each program, not what coverage does. The ER gap is far larger: 95% intervals of 29.4-34.4 against 14.8-17.1 visits per 100 people. Frequent ER users (3+ visits) are 1.3% of people but generate 24% of visits, and 55% of them have public-only coverage. By income, ER use falls from 28.7 visits per 100 people in the poor group to 15.4 in the high-income group, and the uninsured rate falls from 9.4% to 3.3%. Overall, 68 million ER visits cost $95.8 billion, 3.4% of all spending.

Chronic conditions

CHRONIC CONDITIONS (ADULTS)	SHARE OF ADULTS	MEAN SPENDING	SHARE OF ADULT SPENDING	ER VISITS PER 100	DISCHARGES PER 1,000
0	38.9%	$4,046	16.4%	10.9	38.1
1	22.2%	$7,290	16.8%	16.0	60.6
2	15.0%	$10,829	16.9%	23.6	99.6
3-4	17.9%	$17,511	32.6%	34.7	176.1
5+	6.0%	$28,077	17.4%	67.4	383.5

Adults with cancer, heart disease or arthritis spend 2.6-2.7 times as much as adults without the condition.

CMS findings: prices and payments

Medicare paid $75.1 billion for 4.95 million stays ($90.9 billion from all payers) against $457.6 billion in charges. Sepsis alone (DRG 871) is 11.9% of Medicare payments; 6 diagnosis groups make up a quarter of payments and 83 of 540 make up 80%.

CASE-MIX-ADJUSTED CHARGE QUARTILE	HOSPITALS	CHARGE INDEX	PAYMENT INDEX
Lowest 25%	453	0.48	0.90
2nd	452	0.70	0.90
3rd	452	0.96	1.00
Highest 25%	452	1.68	1.03

Across 1,809 hospitals with 500+ discharges, the payment index runs from 0.74 (10th percentile) to 1.22 (90th). Safety-net and academic hospitals dominate the top of the ranking. The state payment index spans 0.76 (Arkansas) to 1.40 (Maryland), a 1.85-fold range, while the charge index spans 5.6-fold (Maryland 0.32 to Nevada 1.77). After case-mix adjustment, metropolitan hospitals sit at 1.01 and rural and micropolitan hospitals at 0.89-0.90.

For sepsis with a major complication (DRG 871, 2,661 hospitals), the 10th-percentile hospital charges $32,161 and the 90th $140,999, a 4.4-fold gap, while payments run from $13,355 to $23,576, 1.8-fold.

Cross-check between the sources

MEPS estimates 12.9 million hospital stays for people 65+, against 5.0 million Medicare fee-for-service stays in the CMS file (38%), which is plausible because the CMS file excludes Medicare Advantage. Cost per stay is the same order of magnitude: $16,784 in MEPS against $18,360 total and $15,166 Medicare-only in CMS.

Policy implications

These are readings of descriptive data, not tested interventions.

1. **Target the high-need minority.** With two-thirds of spending in 10% of people, and 3-4 chronic conditions marking a third of adult spending, care management is worth more per enrollee when aimed at multi-condition adults than at the broad population.
2. **Do not assume coverage drives ER use.** The uninsured use the ER least. Public-only enrollees, who are older and sicker, use it most, and 3+ visit users are 1.3% of people but 24% of visits, so a frequent-user program has a defined target.
3. **Judge hospitals on payment, not charges.** List prices vary 4-fold within a diagnosis but payment only 1.7-1.8-fold, and the top-charging quarter of hospitals is paid 15% more, so charge-based comparisons mislead price transparency work.
4. **Read payment variation with case mix and mission in mind.** The highest payment indexes cluster among safety-net and academic hospitals, and the 1.85-fold range across states is a prompt to ask why, not evidence of inefficiency.

Where AI fits

This analysis is descriptive and uses no machine learning. AI is used on the same problems in five ways, and the first three could extend this project.[1]

1. **Predicting high-cost people.** Gradient-boosted trees or regularized regression predict next year's spending from diagnoses, prior use and demographics, so care management can start with the few people who drive half the spending.
2. **Flagging avoidable use.** Models score the risk of readmission, repeat ER visits or missed appointments so an intervention can reach the patient first.
3. **Finding unusual prices and billing.** Anomaly detection and clustering on claims and charge data surface outlier hospitals or providers and catch coding errors or fraud.
4. **Reading unstructured data.** Language models pull diagnoses and social needs from clinical notes and suggest billing codes.
5. **Cutting administrative work.** Prior authorization, claims processing, scheduling and documentation are the most immediate, least contested uses.

Cautions. A model that finds high-cost patients does not show that any intervention lowers their cost. Models inherit the biases in their data: one widely used care-management algorithm used past cost as a proxy for need and under-identified Black patients who were equally sick (Obermeyer et al., Science, 2019). Evidence of realized savings is thinner than vendor claims, so any model needs a real evaluation.

[1] AI also assisted with this project: it helped write and check the SQL and the drafts, and a person validated the numbers. ↩

Code and scripts

Each script opens with its business question, the SQL skills it shows and its definitions. Output for every query is saved as a CSV in `results/`.

#	SCRIPT	KEY TECHNIQUES
01	`meps_data_quality`	validation queries, weighted reconciliation
02	`meps_spending_concentration`	weighted running totals (`SUM() OVER` with `ROWS` frame), weighted percentile tiers
03	`meps_spending_by_insurance_and_age`	weighted means and shares, conditional aggregation, source of payment
04	`meps_emergency_room_use`	weighted per-capita rates, domain estimation, frequency bands
05	`meps_inpatient_and_service_mix`	composition analysis, ratio-of-sums estimators, `UNION ALL`
06	`meps_equity_and_chronic_conditions`	stratified comparisons, index vs reference group
11	`cms_data_quality`	profiling, anti-joins, logical consistency checks
12	`cms_charges_vs_payments`	discharge-weighted averages, ratio analysis, deciles
13	`cms_drg_concentration`	Pareto analysis, running totals, cumulative share
14	`cms_hospital_benchmarking`	indirect standardization, multi-CTE pipelines, `RANK()` , `NTILE()`
15	`cms_geographic_variation`	state-level standardization, rural-urban comparison
16	`cms_within_drg_variation`	hand-built percentiles, top-N per group, p90/p10 ratios
90	`cross_source_context`	combining two sources, unit reconciliation

To rebuild everything from the public files:

```
pip install -r requirements.txt
python3 download_data.py   # MEPS HC-256 and CMS 2024 CSV into raw_data/
python3 build_db.py        # builds health_policy_real.db
python3 run_all.py         # runs sql/01..90, writes results/
python3 survey_se.py       # standard errors and 95% CIs
python3 make_charts.py     # regenerates images/
```

A clean rebuild in a fresh directory reproduced every result file exactly.

Limitations

- **CMS covers Medicare fee-for-service inpatient stays only.** It excludes Medicare Advantage, Medicaid, commercial insurance and outpatient care, and its payment figure is the hospital facility payment without separately billed physician fees.
- **MEPS is self-reported.** It covers the civilian noninstitutionalized population, so nursing-home residents are out, and small groups carry wide intervals (the uninsured ER rate has a 13% relative standard error).
- **Group differences are unadjusted.** Insurance, race and income comparisons do not control for age, health or each other, and none is causal.
- **The sources are not joined.** The cross-check is a plausibility test, not a reconciliation.
- **Top-tier confidence intervals are slightly narrow.** They treat the spending cutoff as fixed. The 2021 top-1% benchmark (24.0%) falls inside the 2024 interval (19.0-25.9%), so the small difference is not distinguishable from sampling error.
- **Some hospitals are extreme.** Kings County Hospital Center in New York has a payment index of 3.69. The source rows confirm it (sepsis total payment $68,676 against a national

$18,230); this file cannot show why.

Synthetic demo

The repository also holds a reproducible synthetic version: a 12-hospital, 3-year database with 11 scripts covering utilization trends, payer margin, readmissions, ED use, equity and a policy scenario. Its numbers show method only and are not findings. One of its built-in patterns is contradicted by real data: it gives uninsured and self-pay patients more ER use, but in MEPS the uninsured use the ER least.

Sources

- MEPS HC-256, 2024 Full-Year Consolidated Data File (AHRQ)
- Medicare Inpatient Hospitals by Provider and Service (CMS)
- MEPS Statistical Brief #556: concentration of health expenditures, 2021 (AHRQ)
- Obermeyer et al., Dissecting racial bias in an algorithm used to manage the health of populations (Science, 2019)

Case study doc, code and results are in the project repository (`real_data/` and `synthetic_demo/`).